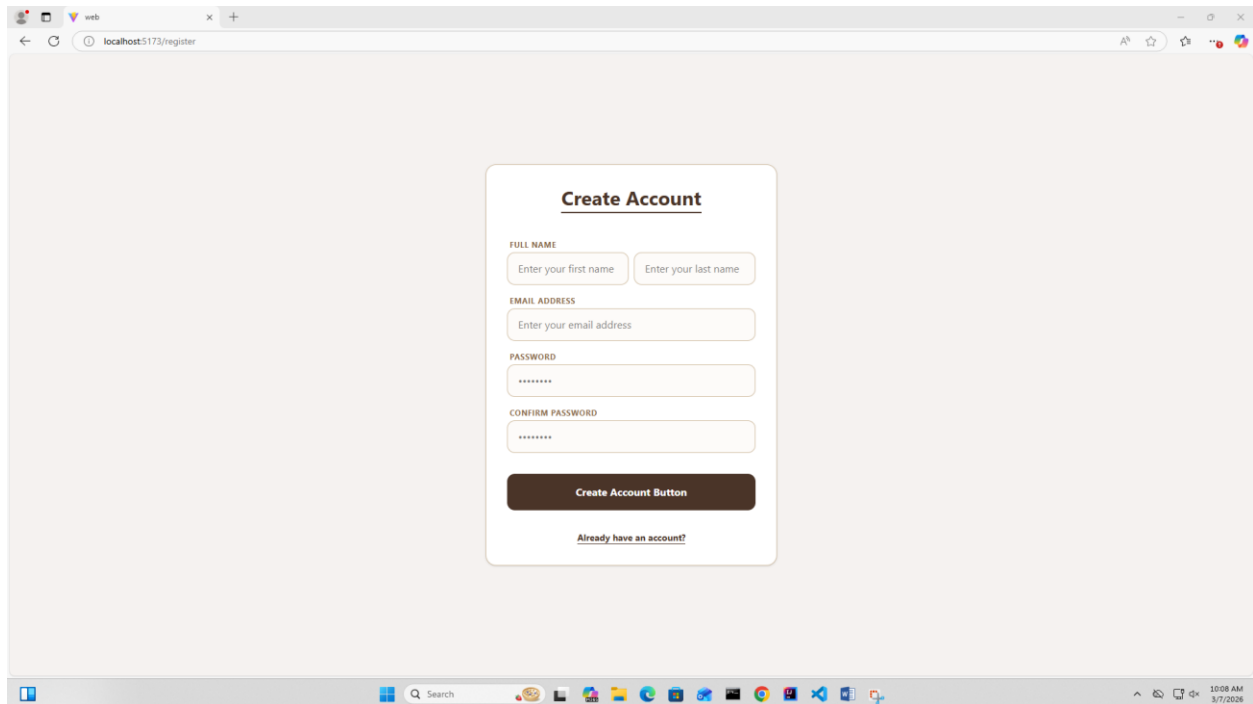
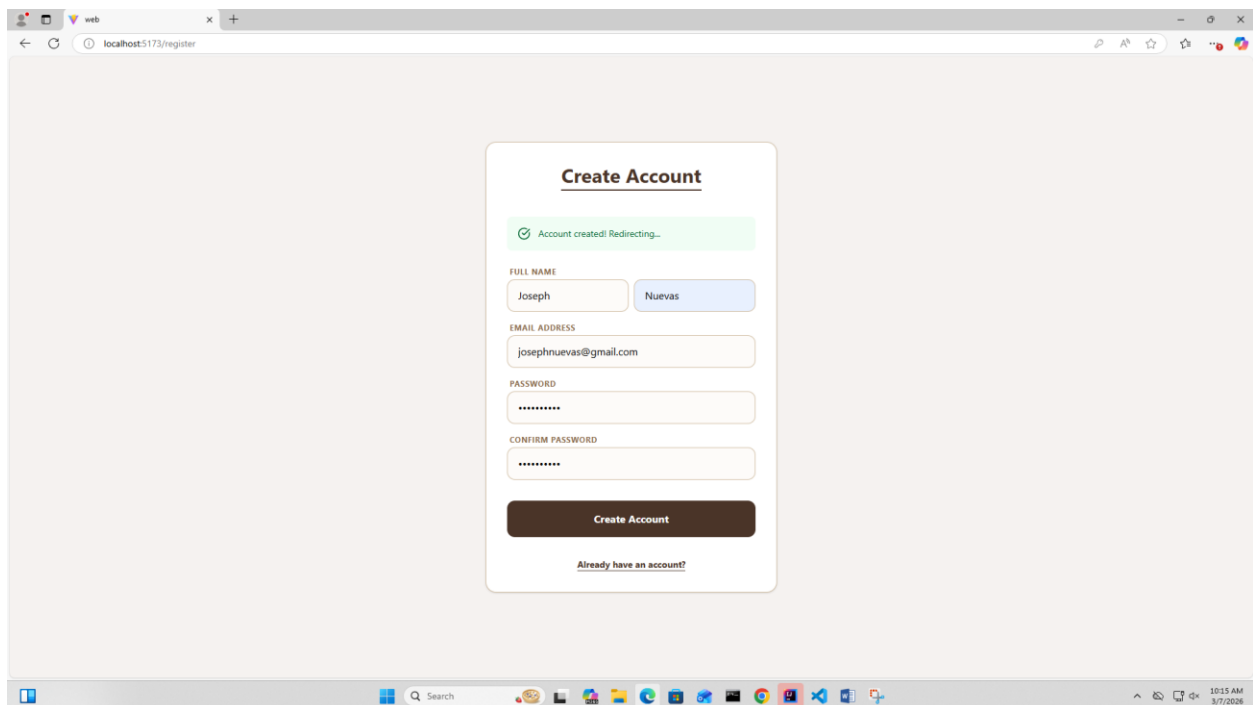


Registration Page



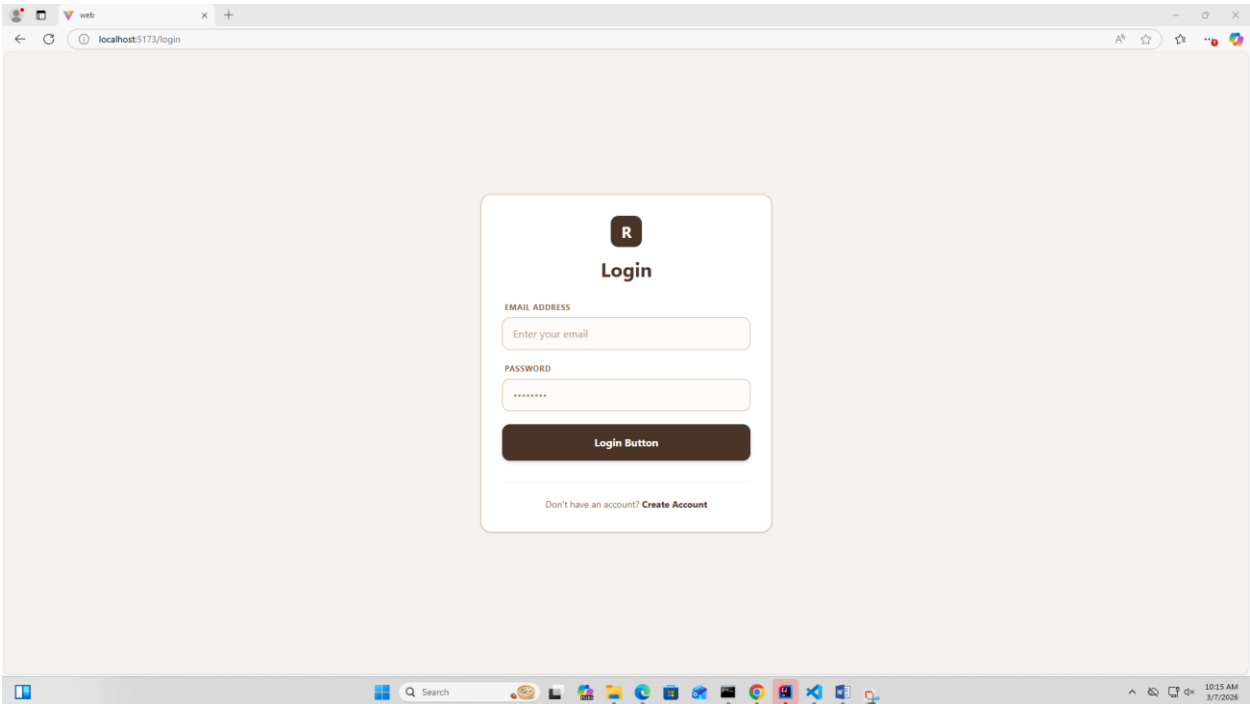
A screenshot of a web browser displaying a registration page at localhost:5173/register. The page features a central 'Create Account' form with the following fields: 'FULL NAME' (split into 'Enter your first name' and 'Enter your last name'), 'EMAIL ADDRESS' (Enter your email address), 'PASSWORD' (masked with dots), and 'CONFIRM PASSWORD' (masked with dots). A dark 'Create Account Button' is at the bottom of the form, with a link 'Already have an account?' below it. The browser's taskbar at the bottom shows various application icons and the system clock at 10:08 AM on 3/7/2026.

Successful User Registration

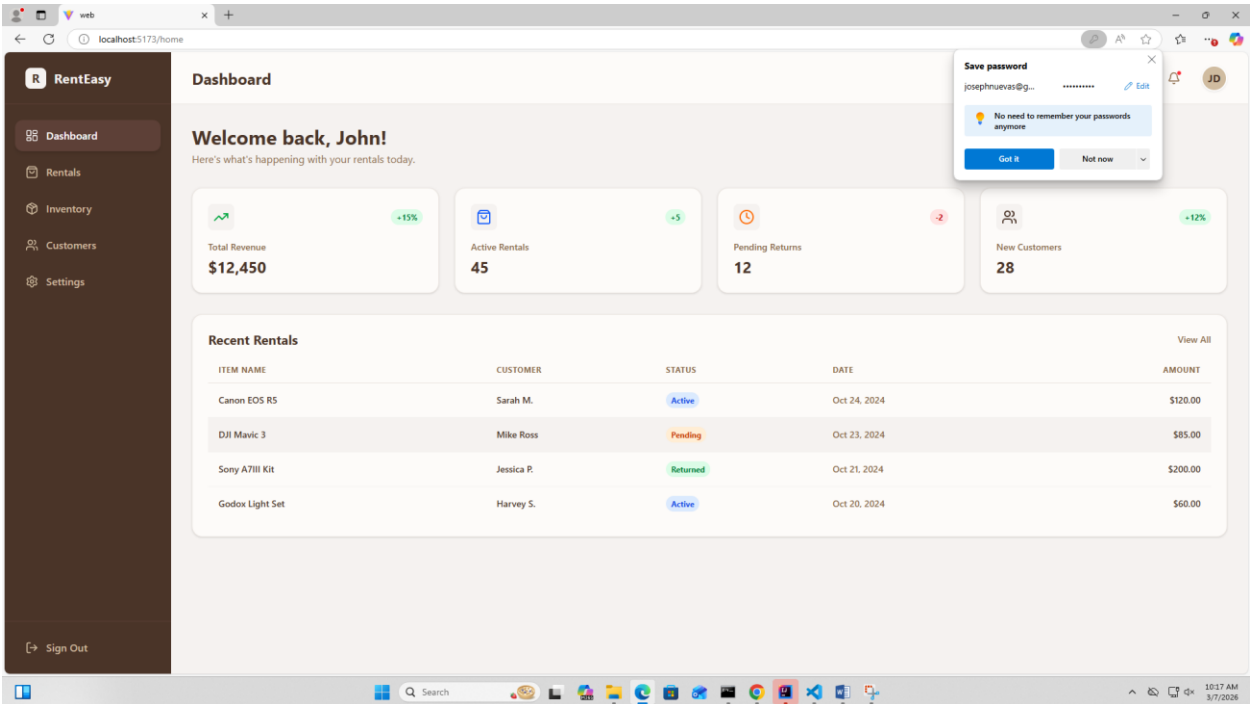


A screenshot of the same registration page after a successful registration. A green success message 'Account created! Redirecting...' with a checkmark icon is displayed at the top of the form. The 'FULL NAME' fields are now populated with 'Joseph' and 'Nuevas'. The 'EMAIL ADDRESS' field contains 'josephnuevas@gmail.com'. The 'PASSWORD' and 'CONFIRM PASSWORD' fields remain masked. The 'Create Account Button' is still present, and the 'Already have an account?' link is at the bottom. The browser's taskbar shows the system clock at 10:15 AM on 3/7/2026.

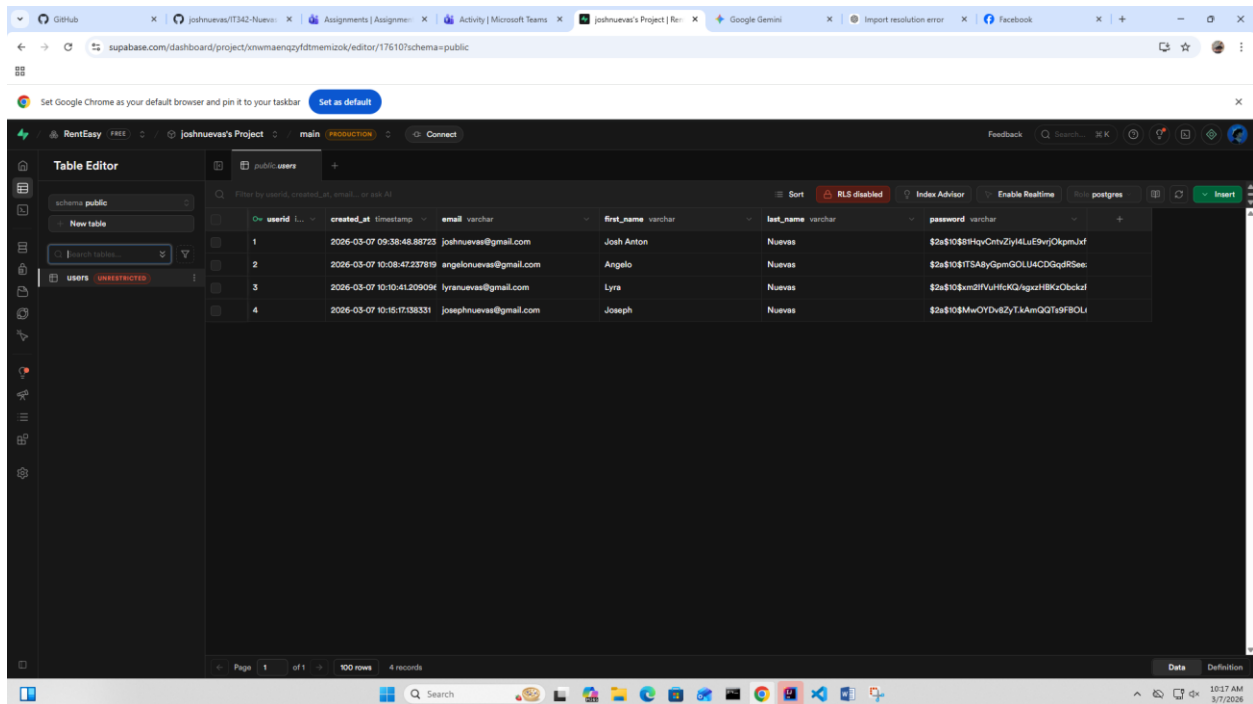
Login Page



Successful Login



Database Record Showing the Registered User



The screenshot displays the Supabase Table Editor interface for the 'users' table in the 'public' schema. The table contains 4 records. The interface includes a sidebar with 'Table Editor' and 'users' table selected, and a top bar with 'RentEasy' and 'main' tabs.

id	username	created_at	email	first_name	last_name	password
1		2026-03-07 09:38:48.88723	joshnuevas@gmail.com	Josh Anton	Nuevas	\$2a\$10\$9HqyCmhzZyM4LuE9vjOkpuxf
2		2026-03-07 10:08:47.237819	angelonuevas@gmail.com	Angelo	Nuevas	\$2a\$10\$ITSaByGpmGOLU4CDGqdrSee
3		2026-03-07 10:10:41.209094	lyranuevas@gmail.com	Lyra	Nuevas	\$2a\$10\$am2fVuhfcKQAgxzHBKzObckuf
4		2026-03-07 10:15:17.138331	josephnuevas@gmail.com	Joseph	Nuevas	\$2a\$10\$MwCYDv8ZyTAAmQQT19fBOLu

Final Commit: <https://github.com/joshnuevas/IT342-Nuevas-RentEasy/commit/6c32d78093b8219d5b96b0fe02342a62ae141867>

User Registration

The registration process allows new users to create an account to start their rental journey.

- **Registration Fields Used:** The system collects the user's firstName, lastName, email, and password.
- **Validation Process:** Basic field requirements are enforced on the frontend using HTML5 validation. On the backend, the system ensures the email is not already in use before proceeding.
- **How Duplicate Accounts are Prevented:** The UserService performs a lookup in the database using findByEmail. If the email is already present, the registration is aborted with an "Email already exists" message.
- **How Passwords are Stored Securely:** Passwords are never stored in plain text. The system uses **BCrypt hashing** to encrypt the password before saving it to the database.

User Login

Registered users can access the system by providing their verified credentials.

- **Login Credentials Used:** Users log in using their registered email and password.
- **How the System Verifies Users:** The system fetches the user by email and uses passwordEncoder.matches() to compare the provided raw password against the hashed version stored in the database.
- **What Happens After Successful Login:** Upon successful verification, the backend generates a **JWT (JSON Web Token)**, which is returned to the client to authorize future requests.

Database Table

The system uses a PostgreSQL table hosted on Supabase to persist user data.

- **Table:** users
- **Columns:** userid (PK), email (Unique), password, first_name, last_name, created_at

API Endpoints

The backend exposes the following REST API endpoints for authentication:

- POST /api/auth/register: Handles new user account creation.
- POST /api/auth/login: Handles user verification and token generation.